

AR²O Painter: An Artistic Oriented Realtime Realistic Oil Painting Agent Powered by Efficient Fluid Simulation

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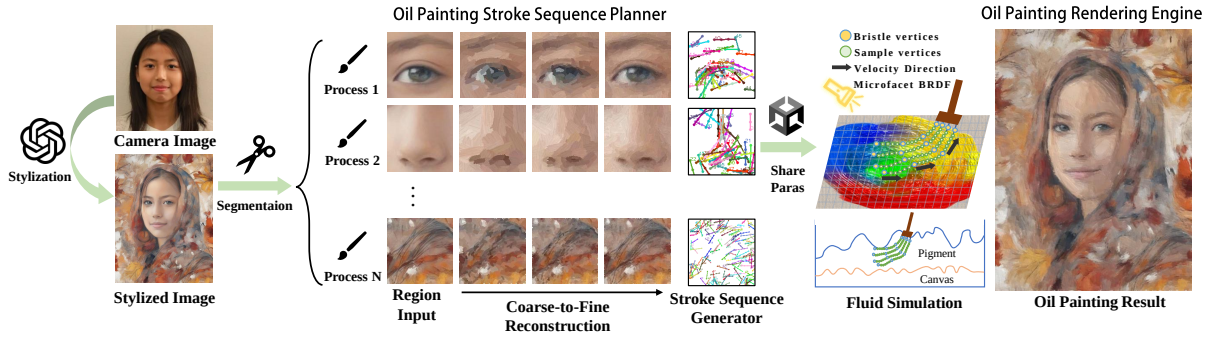


Figure 1: Oil Painting Agent framework. The system captures user’s portrait image and transforms it into an oil painting style. After semantic segmentation, Oil Painting Stroke Sequence Planner starts hierarchical stroke sequence planning, which emulates the artist’s process of progressively refining the painting, from coarse strokes to finer details. The Painting Engine models pigment behavior using fluid dynamics, simulating interactions with the canvas and brush, and utilizes a tailored PBR rendering framework for plausible gloss and surface detail.

Abstract

We introduce the AR²O Painter, an interactive intelligent system designed for real-time, highly realistic oil painting creation. This Agent can faithfully reproduce any portrait image, allowing users to visually enjoy the stroke-by-stroke painting process immersively as the artwork is completed within two minutes. It consists of two modules: the Oil Painting Stroke Sequence Planner, which performs multi-level semantic-based brushstroke sequence decomposition on portrait images, mimicking the logic of artist painting, and the Oil Painting Rendering Engine, which receives the brushstroke sequence, models the pigment via fluid dynamics, simulates its interaction with the canvas and brush, and applies a tailored PBR model with microfacet BRDF, Fresnel effects, and stroke-level geometry, enabling perceptually plausible gloss and fine-grained surface

relief. To the best of our knowledge, it is the first real-time intelligent painting system to generate realistic oil paintings with high interactivity and artistic fidelity. The demo video is available at <https://youtu.be/aN-W06GmnP8>.

CCS Concepts

• **Computing methodologies** → **Modeling and simulation**; • **Applied computing** → **Media arts**.

Keywords

Real-time Realistic Oil Painting, Stroke Based Rendering, Fluid Dynamics, Physically Based Rendering

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1 Introduction

With increasing demand in interactive and personalized art, digital oil painting holds significant application value. Traditional painting is limited by complex materials and skill-intensive processes. While

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digital oil painting alleviates the material constraints, it often lacks realistic paint textures and the logic of artist brushwork. However, current learning based style transfer methods [10, 12], despite their efficiency in capturing color and composition patterns, fundamentally fail to reproduce authentic brushstroke textures due to their pixel-level nature. Most pigment simulations rely on 2D textures, leading to discontinuities and weak brushstroke interactions [14]. While procedural textures [9, 16] improve efficiency at the expense of realism, simulations for brush and fluid [3–7, 18] offer higher accuracy but remain impractical for real-time applications. For brushstroke planning, existing algorithms exhibit several shortcomings, including reconstruction artifacts [12, 13], disordered or redundant brushstrokes [10, 17], and high computational costs [19] due to prolonged decomposition times.

In this work, we address these challenges by introducing our novel digital oil painting system. We simulate pigment behavior with efficient fluid dynamics and render realistic oil painting effects using a PBR render engine. As for brushstroke sequence decomposition, we propose a hierarchical brushstroke planning framework which produces fine-grained stroke sequences, ensuring logical and aesthetically pleasing progression. To enable personalized artwork generation, we incorporate a pretrained stylization module that transforms user portraits with artistic abstraction.

2 System Framework

AR²O Agent’s workflow, as illustrated in Figure 1, features real-time portrait-to-painting transformation supporting arbitrary image inputs with two core modules: Stroke Sequence Planner and Painting Engine. The camera captures a user’s portrait, which is then stylized and processed by Stroke Sequence Planner to perform multi-level semantic brushstroke sequence decomposition, capturing the process of painting of an artist. Simultaneously, Painting Engine receives brushstroke parameters via share memory in a streaming manner to model pigment behavior using fluid dynamics, and render pigment within a tailored PBR rendering framework.

2.1 Oil Painting Stroke Sequence Planner

Stroke Sequence Planner is designed to plan brushstroke sequences that mimic artist painting logic. We implement Hu’s method [11], which employs a hierarchical stroke search with pixel-wise tangential alignment reconstructs painterly stroke sequences within segmented semantic regions. The input image is first divided into regions for parallel processing, where each performs multi-scale stroke search by progressively reducing filter kernels along texture gradient tangents, dynamically adjusting stroke size to achieve pixel-wise matching to the input image. InsightFace [8] is employed for precise facial segmentation, while a complementary grid-based segmentation is implemented for background regions.

2.2 Oil Painting Rendering Engine

The Oil Painting Rendering Engine constructs a deformable 3D brush model using a position dynamics system, stabilized by a two-stage driving mechanism with length and curvature constraints to simulate realistic elastic forces. Pigment is stored on the subdivided vertices of the bristles and applied to the canvas using a fast Splat-based method, which leverages GPU’s triangle rasterization for

high-speed rendering. To enable efficient pigment simulation, the system adopts a lightweight 2D grid-based fluid dynamics model, which captures realistic interactions among pigment, brush and canvas with minimal computational cost. The canvas is also simulated to support physically plausible pigment mixing and absorption. Rendering is handled through a custom PBR framework, integrating a microfacet BRDF with Fresnel effects for realistic gloss and specular detail. Stroke-level geometry modeling further enhances the surface fidelity of brushwork.

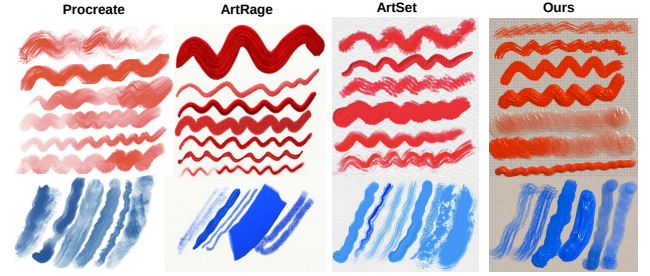


Figure 2: Qualitative Comparison of generated stroke quality.

3 Experiment Results

We perform both quantitative and qualitative evaluations at a workstation with Intel i9-14900KF and RTX 4090. The quantitative experiment presents the comparison of the reconstruction metrics between Stroke Sequence Planner and other SBR methods, demonstrating the superiority of our method in both quality and speed, as shown in Table 1. For the qualitative evaluation in Table 2, we recruit 30 volunteers to rate the stroke generation quality of the Rendering Engine against existing painting software, as illustrated in Figure 2. The score results highlight the superior realism and physicality of our oil painting strokes. In particular, in terms of stroke glossiness and the tactile texture of the brush bristles within the strokes, our results have received great praise from users. Note that, unlike existing digital painting software that requires users to instruct manual drawing, our system uniquely incorporates the Paint Agent to intelligently plan sequential brushstrokes.

Table 1: Quantitative Comparison with SBR SOTAs.

Methods	PSNR↑	SSIM↑	LPIPS↓	Search Time↓	Ranking↓	Platform
Huang [12]	28.58	0.7338	0.3182	10.3 ms/stroke	4.83	GPU
Hu [10]	28.46	0.6878	0.2884	7.7 ms/stroke	2.17	GPU
Liu [13]	24.46	0.6132	0.3955	25.4 ms/stroke	4.97	GPU
Zou [19]	20.87	0.4906	0.4330	358.1 ms/stroke	5.20	GPU
Tong [17]	28.54	0.7092	0.2042	30.5 ms/stroke	2.33	CPU
Ours	28.98	0.7714	0.1873	7.1 ms/stroke	1.50	CPU

Table 2: User Study on stroke assessment with painting software. The score ranges 1-5.

Methods	Color↑	Thickness↑	Dynamics↑	Light↑	Realism↑	PaintAgent
Procreate [15]	4.27	2.23	3.20	2.20	3.20	✗
ArtRage [2]	4.13	3.87	3.67	3.13	3.70	✗
ArtSet4 [1]	4.23	3.13	3.40	2.93	3.83	✗
Ours	4.57	4.47	4.23	4.60	4.73	✓

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